

EOL Manufacturing Execution System Setup, Operations & Maintenance Guide

Toyota Boshoku Device India | YNC-SS Seat Slider Line | MES Phase 2 + Camera CMS

Audience: Manufacturing/automation engineers, maintenance technicians, IT operators, plant managers, and anyone who needs to stand up, operate, or troubleshoot the EOL stack.

Scope: Complete cradle-to-grave reference covering hardware topology, first-time installation, daily operations, breakdown / CAPA workflow, troubleshooting playbooks, and architecture deep-dive.

Document version

v1.0 • 2026-05-20 • Generated from live production system

Contents

§	Section
1	System Overview & Architecture
2	Hardware Topology
3	First-Time Installation
4	Daily Operations (start / stop / verify)
5	Service Reference (ports, endpoints, processes)
6	Database Schema Reference
7	Maintenance Workflows (breakdown slip, CAPA, escalation)
8	Shift Configuration
9	Camera & Video Pipeline
10	Troubleshooting Playbook (top 12 known issues)
11	GPU & Encoding (NVENC / QSV / libx264)
12	Backup & Disaster Recovery
13	Security Notes
14	Glossary

1. System Overview & Architecture

EOL = End-of-Line inspection. Two cooperating apps:

1.1 MES (Manufacturing Execution System)

FastAPI backend (port 8080) + React frontend (port 5656). Polls the Mitsubishi PLC every 30 ms, records every OK / NG cycle to PostgreSQL, serves the operator dashboard, drives the breakdown / CAPA workflow.

1.2 CMS (Camera Management System)

Flask api_server (port 5555) + React frontend (port 5575). Runs one ffmpeg process per IP camera, recording H.265 RTSP streams to disk 24/7. On every cycle end, extracts a per-cycle MP4 clip indexed by part_code so operators can replay any cycle from the dashboard.

1.3 Why two apps?

Historical separation — MES (Phase 2) was built first. CMS came later when video recording was added. They share the PLC (both poll independently at 30 ms) and exchange cycle-end events via a localhost HTTP webhook (/api/plc-edge). Future consolidation possible but not blocking.

1.4 Port + service quick map

Port	Service	Tech	Started by
8080	MES Backend	Python / FastAPI / uvicorn	start_everything.bat [1/5]
—	MES Collector	Python / pymcprotocol	start_everything.bat [2/5]
5656	MES Frontend	Node / Vite	start_everything.bat [3/5]
5555	CMS API	Python / Flask + plc_monitor	start_everything.bat [4/5]
5575	CMS Frontend	Node / Vite	start_everything.bat [5/5]

2. Hardware Topology

Device	Spec	Network address
MES Server	HP Z2 Tower G9 Workstation, RTX A2000 12 GB GPU, 8 CPU cores, 32 GB RAM, D: 931 GB SSD + F: 3.7 TB external HDD	TBDI - BI - SVR - 01
PostgreSQL	PostgreSQL 16 (energydb database)	192.168.10.210 : 5432
PLC	Mitsubishi Q-series, MC4E protocol, polled @ 30 ms	192.168.10.150 : 5002
Camera 1	Panasonic i-PRO PI-BUA3MWA 3MP (Final Inspection)	192.168.10.115 : 554 RTSP
Camera 2	Sub-machine bullet (Ball Guide)	192.168.10.127
Camera 3	Sub-machine bullet (Lock Bar)	192.168.10.118
Camera 4	Sub-machine bullet (Lock Bar Insert)	192.168.10.156
Camera 5	Sub-machine bullet (Semi-Auto)	192.168.10.158
Camera 6	Sub-machine bullet (Upper Rail Greasing)	192.168.10.122
Shop-floor TV	Wall-mounted 16:9 panel, runs MES Frontend full-screen	Browser kiosk

2.1 Key PLC bits

Bit	Meaning
L108	OK / cycle-end pulse (~50-100 ms wide)
L109	NG / quality reject pulse
D6005	Status word — bit 4 = remote-active
M157	Semi-Auto cycle bit (pending PLC engineer ladder confirm)

3. First-Time Installation

Use this section when standing up a fresh server.

3.1 Prerequisites

Item	Version / Notes
Windows	Windows 11 Pro or Windows Server 2022
Python	3.12.x (add to PATH during install)
Node.js	20.x LTS (bundles npm)
Git	2.40+ (Git for Windows)
PostgreSQL client	psql / pgAdmin (for DB checks)
NVIDIA driver	≥ 551.76 (Production Branch R580 recommended)
Disk space	200 GB system + 4 TB external (video storage)
Network	Static IPs for PLC + cameras, port 5432 to DB host

3.2 Clone and configure

```
git clone https://github.com/<your-id>/EOL.git D:\EOL\EOL
cd D:\EOL\EOL

:: MES backend
cd "Deep (2)\Deep\Phase2"
python -m venv .env
.venv\Scripts\activate
pip install -r requirements.txt
copy .env.example .env
:: Edit .env with your DB credentials (DB_HOST, DB_PASS, etc.)

:: MES frontend
cd ..\..\..\mes-frontend
npm install

:: CMS backend
cd ..\..\New folder (2)\New folder (2)\backend
pip install -r requirements.txt
copy .env.example .env

:: CMS frontend
cd ..\frontend
npm install
```

3.3 Database init

Tables are created lazily by the backends on first call. Seed data (zones, lines, shifts, machines) must be inserted manually. Use pgAdmin or psql:

```
psql -h 192.168.10.210 -U postgres -d energydb

-- Seed a zone, line, and shift config:
INSERT INTO mes_zones (zone_code, zone_name, is_active)
VALUES ('ZONE-1', 'SEAT SLIDER', TRUE);

INSERT INTO mes_lines (line_code, line_name, zone_id, db_table_name,
                      ideal_cycle_time, is_active)
VALUES ('YNC-L6', 'YNC-SS', 1,
        'ync_dashboard_complete', 15.0, TRUE);
```

3.4 First launch

```
cd D:\EOL\EOL\Deep (2)\Deep
start_everything.bat
```

Open <http://127.0.0.1:5656> — default login is admin / admin123. Change the password on first login.

4. Daily Operations

4.1 Start the system (shift begin)

```
cd D:\EOL\EOL\Deep (2)\Deep
start_everything.bat
```

Launcher opens 5 service windows over ~40 sec. Each window shows -> *launched*. Waiting *Ns* for port :NNNN to bind... followed by -> *warmed*. If a step hangs > 30 sec, kill the launcher and check that Defender / antivirus isn't blocking process creation.

4.2 Stop everything (shift end / maintenance)

```
cd D:\EOL\EOL\Deep (2)\Deep
stop_everything.bat
```

4.3 Verify everything came up

From PowerShell:

```
# All four service ports should be LISTENING
Get-NetTCPConnection -LocalPort 8080,5555,5656,5575 -State Listen `
-ErrorAction SilentlyContinue
```

```
# 6 ffmpeg recorder processes (one per camera)
tasklist /FI "IMAGENAME eq ffmpeg-win-x86_64-v7.1.exe" /FO CSV /NH
```

4.4 Operator checklist (start of shift)

#	Check	Pass criterion
1	MES dashboard loads	http://127.0.0.1:5656 shows realtime tile
2	Cycle counter advancing	OK / NG / Plan numbers update within 60 s
3	All cameras green	Sub-machine row shows green dot, not red/yellow
4	PLC connected	Light on Realtime tile is GREEN
5	Shift label correct	Header shows current shift (Shift 1 / Shift 2 / OT)
6	No active breakdown	ANDON section says "all lines running smoothly"

5. Service Reference

5.1 Key MES endpoints

Path	Purpose
GET /api/lines/{id}/realtime	Live counter, status, OEE for one line
GET /api/lines/{id}/wallboard-summary	Aggregate KPIs for wallboard tile
GET /api/lines/{id}/wallboard-cycles	Recent cycles for CT chart
GET /api/lines/{id}/cycle-video?cycle_seq=N	Cycle MP4 (proxies to CMS)
GET /api/submachines/{id}/cycle-video?cycle_seq=N	Sub-machine cycle clip
GET /api/breakdowns/active	OPEN breakdowns (ANDON)
GET /api/breakdowns/recent?days=2	RESOLVED + CLOSED last N days
GET /api/breakdowns/history?state=RESOLVED&days=180	Pending closure tickets
POST /api/breakdowns/{id}/production-fill	Production saves their slip half
POST /api/breakdowns/{id}/close	Maintenance closes ticket
GET /api/capa/pending	Auto-detected CAPAs needing action
GET /api/breakdowns/stats?days=30	MTBF / MTTR / LTTR per zone+line+machine

5.2 Key CMS endpoints

Path	Purpose
GET /api/video/by-part?code=<part_code>	Stream cycle MP4 by part code
GET /api/submachine/clip?plc_ip=...&ts_start=...&ts_end=...	On-demand clip from rolling TS
POST /api/plc-edge	Webhook: MES collector reports L108 edge
GET /api/cameras	Camera config (admin)
POST /api/auth/login	Login (admin)

6. Database Schema (key tables)

Full per-column reference is in *MES_Maintenance_DB_Schema.pdf* (see docs/ folder). Quick map:

Table	Role
mes_zones	Plant zones (e.g., SEAT SLIDER)
mes_lines	Production lines
mes_machines	Sub-machine catalog (~285 rows)
mes_breakdowns	MASTER breakdown table — every breakdown plant-wide lands here, filterable by line_id / zone_id
mes_breakdown_mail_levels	Escalation matrix (Level 1/2/3 contacts)
mes_breakdown_mail_log	Audit: every escalation email actually sent
mes_breakdown_slip_config	Global slip-raise threshold (minutes)
mes_capa	CAPA filings
mes_capa_thresholds	Per-line / per-machine CAPA trigger limits
pm_check_master	Preventive maintenance checklist items
repair_work	Multi-step repair log tied to breakdown
mes_collector_locks	Singleton lock for collector process
mes_submachine_ct_log	Per-sub-machine cycle log (ts_start/ts_end)
ync_dashboard_complete_ct_log	Per-line cycle log (line-specific table)

7. Maintenance Workflows

7.1 Breakdown lifecycle

```
OPEN  -->  RESOLVED  -->  CLOSED
|         |         |
|         |         |
|         |         | +-- Maintenance has filed closure form
|         |         | +-- Line is back running
+-- Line currently down (visible on ANDON)
```

7.2 Slip-raise threshold (default 10 min)

Quick-attended events (under 10 min) are MINOR — no formal slip needed. Anything over 10 min is MAJOR — full slip mandatory. Threshold tunable from Maintenance Panel → Slip Threshold.

7.3 The slip itself (Break Down Slip)

Field	Filled by	Lockable
B/D Start / End Times, M/C Down Time	System (auto)	Always locked
Zone, Line, Machine, Model, Operator	Production	Locked after save
Category A/B/C	Production	Locked after save
Problem reported by Production	Production	Locked after save
Problem related to (Maintenance / Tool Room)	Maintenance	Locked after save
Type of problem (Electrical / Mechanical)	Maintenance — multi-select	Locked after save
Actual problem observed	Maintenance	Locked after save
Action taken	Maintenance	Locked after save
Spares used	Maintenance	Locked after save
Attended by	Maintenance	Locked after save
Signatures (prepared / received / QE / leader)	Maintenance + Production	Locked after save

7.4 CAPA triggers

CAPA auto-raised when:

- **SINGLE_LIMIT** — one breakdown exceeds *single_breakdown_minutes_limit* for its scope
- **MONTHLY_LIMIT** — month-to-date sum for a (line, machine_no) crosses *monthly_sum_minutes_limit*

Threshold lookup precedence: Machine → Line → Global

8. Shift Configuration

Shifts are defined in New folder (2)/New folder (2)/backend/shifts.json. CMS uses these to label cycle clip storage paths (F:\MES_Videos\<Zone>\<Line>\<Machine>\<Date>\<Shift>\).

8.1 Current configuration (Toyota Boshoku 2-shift)

Shift name	Start	End	Hours
Shift 1	08:30	17:20	8h 50m
OT Lunch Gap	17:20	18:30	1h 10m
Shift 2	18:30	03:20 (next day)	8h 50m
OT Night Gap	03:20	08:30	5h 10m

8.2 Changing shifts

```
{
  "shifts": [
    { "id": "shift_1", "name": "Shift 1",
      "start": "08:30", "end": "17:20" },
    { "id": "shift_2", "name": "Shift 2",
      "start": "18:30", "end": "03:20" }
  ]
}
```

Edit the file — **no service restart needed**. CMS re-reads on every cycle clip extraction.

9. Camera & Video Pipeline

9.1 Recording flow

```
RTSP camera
|
ffmpeg recorder (one per camera, 24/7)
|
v
Rolling TS file on disk
(e.g. D:\...\videos\cam_panasonic_default_NNN.ts)
|
L108 edge fires (cycle end)
|
v
CMS plc_monitor detects edge, after 25 s sleep
spawns ffmpeg EXTRACTOR which trims the cycle
slice from the TS into a per-cycle MP4 file
|
v
videos/<Line>/<part_code>.mp4
|
Dashboard fetches via /api/video/by-part
```

9.2 Storage paths (current)

Recording falls back to D: drive because F:\ exFAT permissions block Medium-IL writes. Long-term fix: reformat F: to NTFS OR
icacls F:\MES_Videos /grant "Users:(OI)(CI)F" /T.

9.3 Shift-boundary cleanup

At each shift boundary (e.g. 06:00 default), CMS rotates all 6 TS files (closes old, starts new) and wipes the previous shift's MP4 files to keep disk usage bounded. Operator should be aware that cycle clips from a past shift are NOT retained.

10. Troubleshooting Playbook

Top 12 known issues seen on this server, with field-tested fixes.

#	Symptom	Root cause	Fix
1	Cycle videos never appear, but TS files keep growing	Zombie-reaper killing cycle extractor ffmpeggs	Already fixed (plc_monitor.py). Reaper now only targets RTSP recorders.
2	All services stop suddenly at random time	Defender / Windows Update process scan killed api_server	Run start_everything.bat again. Consider exclusion of Python from Defender.
3	MES Backend window starts but launcher stalls	timeout /t N command blocked by Defender stdin hook	Already fixed. Launcher uses ping -n N+1 127.0.0.1 now.
4	Operator clicks dot, video shows "No video available" for 5 sec	Race condition — clip extraction still running	Wallboard auto-retries 4× at 5 s spacing — just wait.
5	CMS-API window shows "'_run_collector.bat' is not recognized"	Missing .\ prefix	Already fixed in start_everything.bat: cmd /k ".\run_collector.bat".
6	Collector window shows "The filename, directory name, or volume label syntax is incorrect"	cmd /k nested-quote parsing of "\"path with spaces\""	Already fixed. Collector launched via _run_collector.bat wrapper.
7	"Another collector is already running" error	Singleton lock from a prior dead process is stale	DELETE FROM mes_collector_locks; OR wait 30s for heartbeat timeout.
8	Live recording works but cycle clips fail to encode	NVIDIA driver too old (< 551.76) for NVENC API 12.2	Auto-falls back to QSV (Intel iGPU). Upgrade driver to R580 Production Branch for full speed.
9	Cycle clips silently going to wrong drive (D: instead of F:)	F:\ exFAT permission probe fails for Medium-IL processes	Run icacls grant once (Section 9.2), OR reformat F: to NTFS.
10	Dashboard says "Shift_C" but factory has only 2 shifts	shifts.json had factory-default 3-shift config	Edit shifts.json to match real schedule. No restart needed.
11	Recent Breakdowns table shows old < 10 min events as "pending closure"	Minor events were being shown as if they need slip	Already filtered. Below threshold = MINOR, hidden from prompt list.
12	Panasonic camera shows blank in dashboard	Camera firmware glitches under heavy network load	Watcher at C:\Users\DX-ADMIN\AppData\Local\Templeol_pana_watch monitors log-mtime. Operator can power-cycle camera if stale > 5 min.

11. GPU & Encoding

Cycle clip re-encoding uses a priority chain:

Priority	Encoder	When used	Speed
1	h264_nvenc (NVIDIA A2000)	Driver ≥ 551.76 + NVENC functional	16× realtime, ~0% CPU
2	h264_qsv (Intel UHD 770)	NVENC unavailable, iGPU present	9× realtime, ~5% CPU
3	libx264 (CPU)	Both above unavailable	17× realtime, ~700% CPU

The picker probes encoders once at process startup (in `_pick_hw_encoder()`) and caches the result. Look for `[ENCODER] picked '<codec>' for H.264 re-encoding` in the CMS-API console to confirm which path is active.

12. Backup & Disaster Recovery

12.1 What to back up

Source	Frequency	Destination
PostgreSQL energydb dump	Daily (cron 02:00)	External NAS or cloud (encrypted)
backend/*.json config files	On every change	Git tag or manual zip
shifts.json, cameras.json	On every edit	Git commit
Full system image (Macrium / Windows Image)	Before driver / OS upgrade	External HDD
Camera video archive	Weekly cold copy (optional)	Separate 4 TB external

12.2 Postgres dump command

```
pg_dump -h 192.168.10.210 -U postgres -d energydb -Fc -Z 9 \  
-f "D:\backups\energydb_$(Get-Date -Format 'yyyyMMdd_HH:mm:ss').dump"
```

12.3 Restore

```
pg_restore -h 192.168.10.210 -U postgres -d energydb \  
--clean --if-exists -j 4 "D:\backups\energydb_20260520_023000.dump"
```

13. Security Notes

Concern	Status	Remediation
DB password in source	Hardcoded fallback in <code>database.py</code>	Set <code>DB_PASS</code> env var, remove default
JWT secret in source	Hardcoded in <code>auth.py</code>	Move to env var, rotate
Camera credentials	Encrypted in <code>cameras.json</code>	Fernet key in <code>secret.key</code> — keep off git
HTTPS for dashboards	Not yet (HTTP only)	Add nginx + Let's Encrypt for production
Admin password default	<code>admin / admin123</code>	Change immediately after first login
Firewall / network ACL	Plant LAN only	Confirm router blocks inbound 8080/5555/5656/5575 from WAN

14. Glossary

Term	Meaning
EOL	End-of-Line inspection
MES	Manufacturing Execution System
CMS	Camera Management System (custom in-house, NOT CCTV)
MC4E	Mitsubishi MELSEC Communication, MC protocol type 4E
L108 / L109	Mitsubishi PLC link relays — OK pulse / NG pulse
CT	Cycle Time
OEE	Overall Equipment Effectiveness
MTBF	Mean Time Between Failures
MTTR	Mean Time To Repair (average)
LTTR	Longest Time To Repair (worst-case, used for zone stats)
LTBF	Longest Time Between Failures (longest survival without breakdown)
CAPA	Corrective Action / Preventive Action
ANDON	Live operator alert board
BREAK DOWN SLIP	Toyota Boshoku formal breakdown documentation
NVENC	NVIDIA video encoder hardware block (offloads from CPU)
QSV	Intel Quick Sync Video (iGPU equivalent of NVENC)
RTSP	Real Time Streaming Protocol — used by IP cameras
TS (file)	MPEG Transport Stream — ffmpeg rolling recording format
Wallboard	Shop-floor TV-mounted live status view
Singleton lock	DB row that ensures only one collector polls a given line

End of EOL Setup, Operations & Maintenance Guide. For DB schema reference see MES_Maintenance_DB_Schema.pdf. For traffic flow diagram see MES_Traffic_Flow.png. For workstation BoM see VMS_Workstation_BoM_Final.pdf.